UNIVERSITY OF WAH (UOW)

DEPARTMENT OF COMPUTER SCIENCE

BSCS 3RD B

REG #: UW-24-CS-BS-088

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COURSE TITLE: COMPUTER NETWORKS

COURSE CODE: CS – 241 L

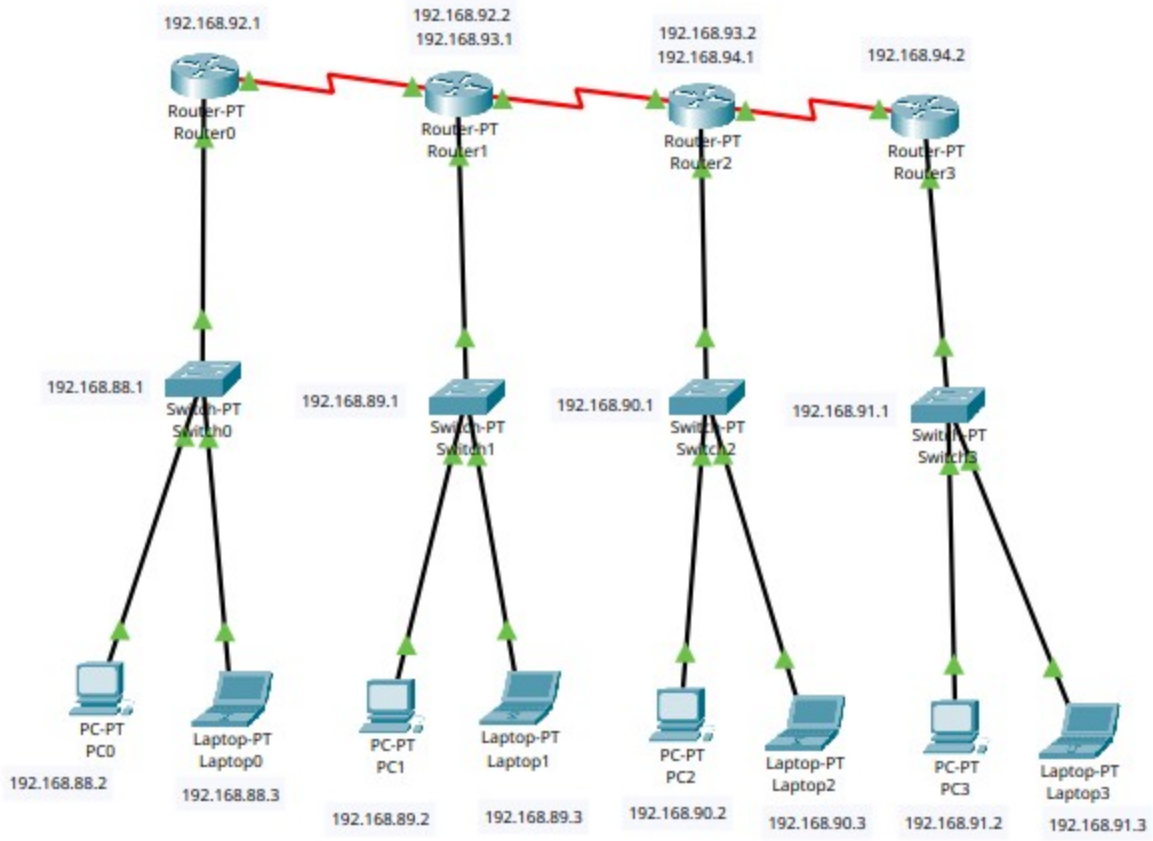
LAB REPORT # 11

1. Objective

The objective of this lab is to simulate dynamic routing using Cisco Packet Tracer. The network consists of four PT routers, four PT switches, and eight end devices. The routers are interconnected using Serial DCE connections, while all other devices use FastEthernet links.

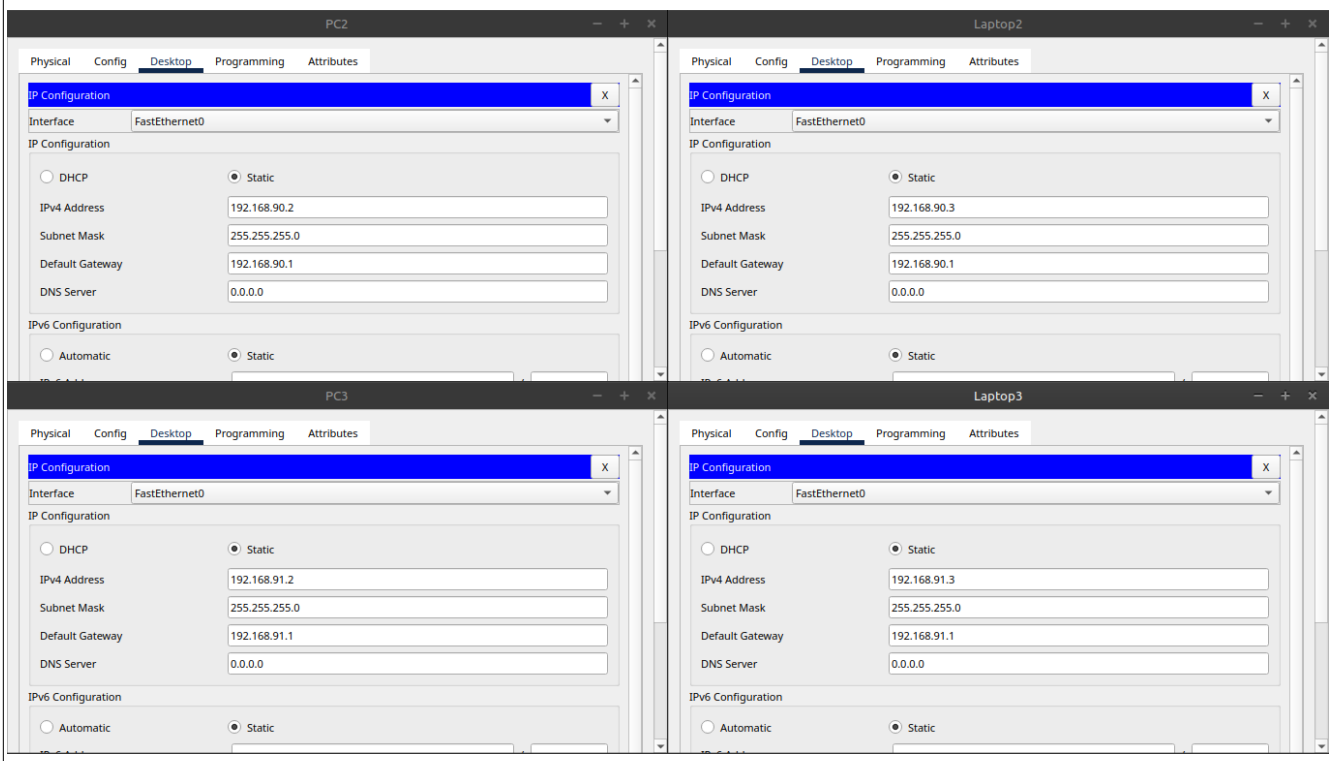
2. Network Topology

The network topology includes:
- 4 Routers connected through Serial DCE interfaces
- 4 Switches connected to the routers
- 8 End devices (PCs) connected to switches



3. IP Addressing

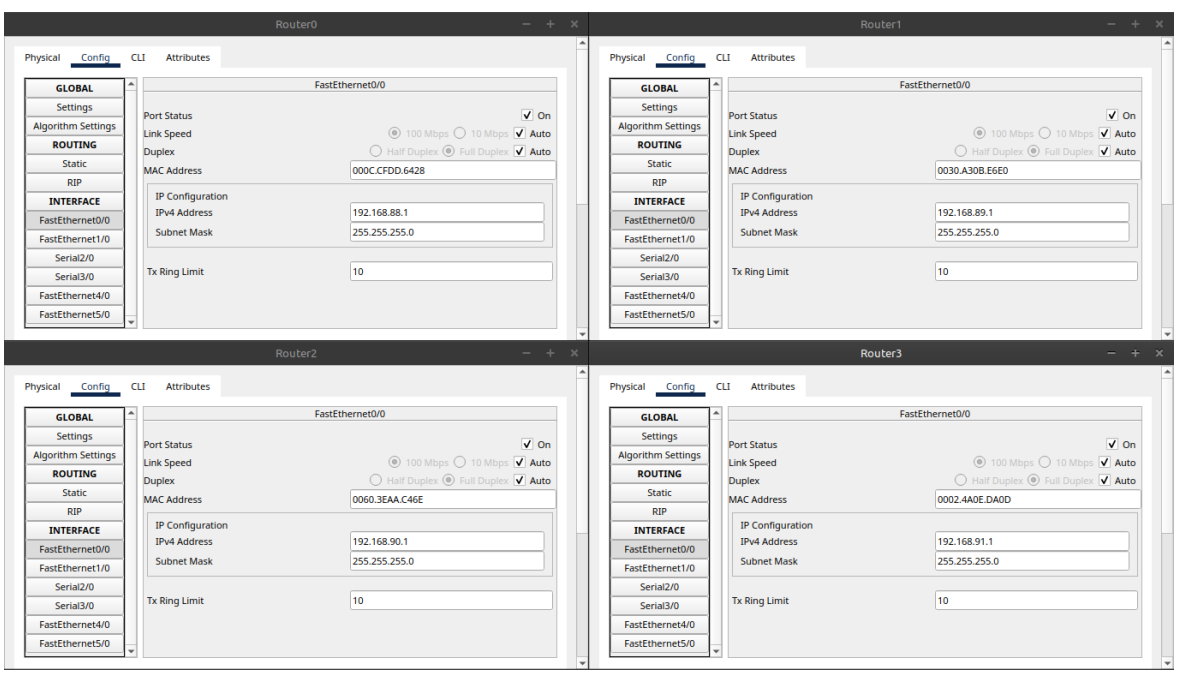
IP addresses were assigned to all end devices, along with their default gateways. Router interfaces, including FastEthernet and Serial ports, were configured with appropriate IP addresses and subnet masks.

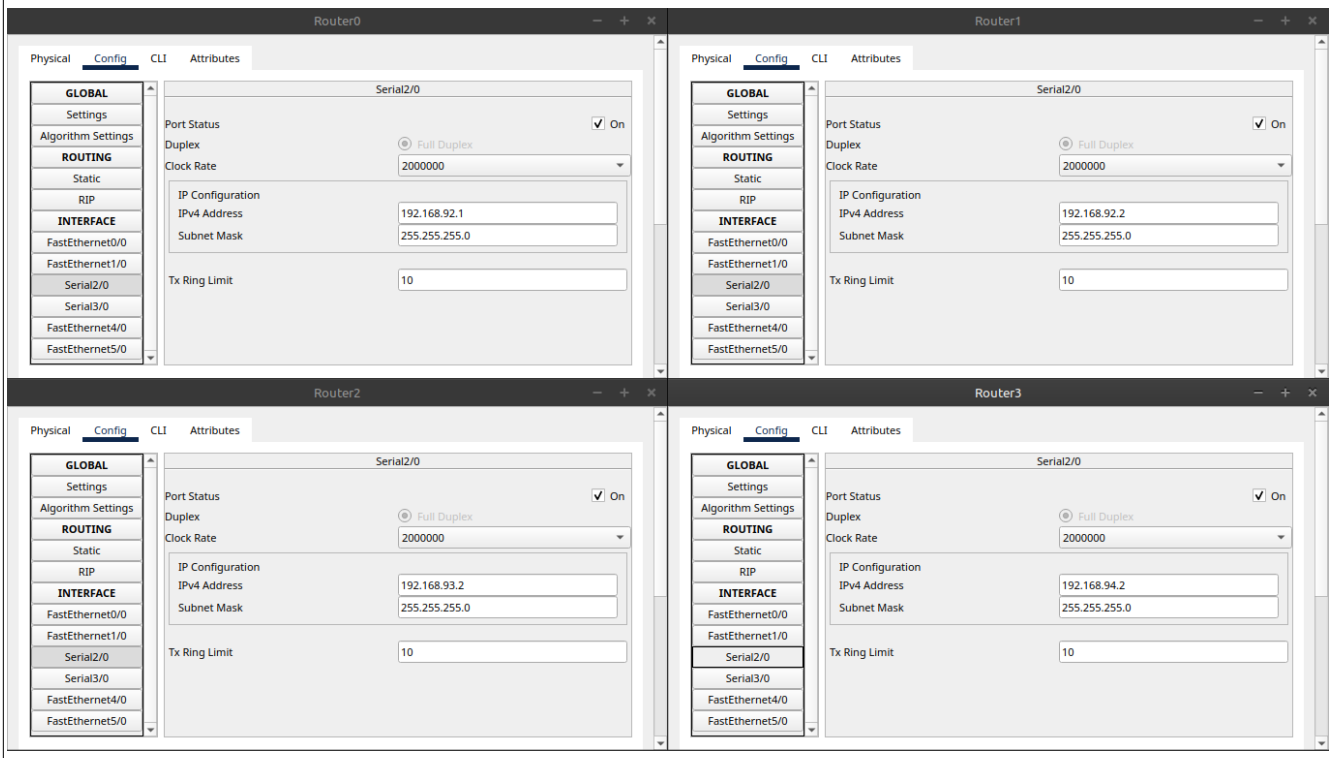


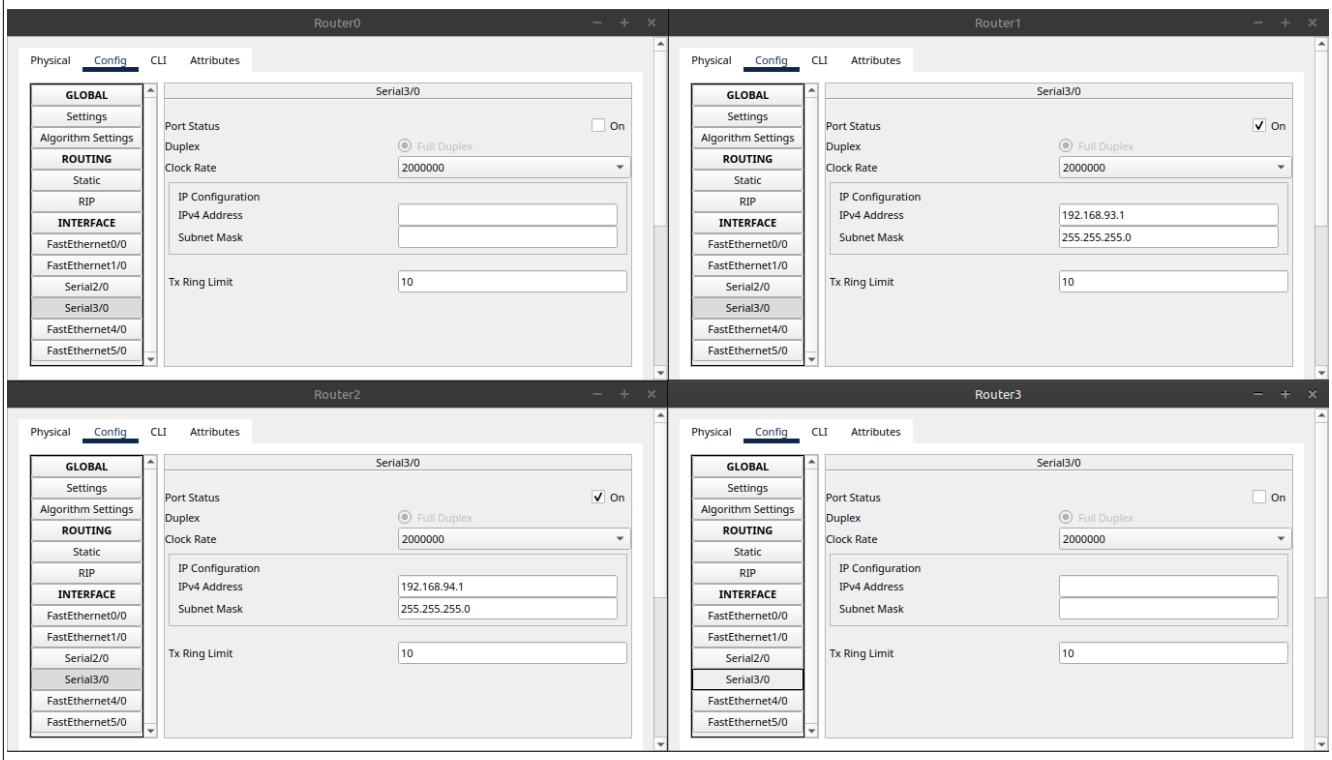
4. Router Configuration

Router configurations included:

- Assigning FastEthernet interface IP addresses
- Setting Serial interface IP addresses for inter-router connectivity
- Enabling RIP dynamic routing and adding network addresses




Router0
Physical Config CLI Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
INTERFACE
FastEthernet0/0
FastEthernet1/0
Serial2/0
Serial3/0
FastEthernet4/0
FastEthernet5/0
Serial2/0
Port Status On
Duplex Full Duplex
Clock Rate 2000000
IP Configuration
IPv4 Address 192.168.92.1
Subnet Mask 255.255.255.0
Tx Ring Limit 10
Router1
Physical Config CLI Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
INTERFACE
FastEthernet0/0
FastEthernet1/0
Serial2/0
Serial3/0
FastEthernet4/0
FastEthernet5/0
Serial2/0
Port Status On
Duplex Full Duplex
Clock Rate 2000000
IP Configuration
IPv4 Address 192.168.92.2
Subnet Mask 255.255.255.0
Tx Ring Limit 10
Router2
Physical Config CLI Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
INTERFACE
FastEthernet0/0
FastEthernet1/0
Serial2/0
Serial3/0
FastEthernet4/0
FastEthernet5/0
Serial2/0
Port Status On
Duplex Full Duplex
Clock Rate 2000000
IP Configuration
IPv4 Address 192.168.93.2
Subnet Mask 255.255.255.0
Tx Ring Limit 10
Router3
Physical Config CLI Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
INTERFACE
FastEthernet0/0
FastEthernet1/0
Serial2/0
Serial3/0
FastEthernet4/0
FastEthernet5/0
Serial2/0
Port Status On
Duplex Full Duplex
Clock Rate 2000000
IP Configuration
IPv4 Address 192.168.94.2
Subnet Mask 255.255.255.0
Tx Ring Limit 10


Router0
Physical Config CLI Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
INTERFACE
FastEthernet0/0
FastEthernet1/0
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IPv4 Address
Subnet Mask
Tx Ring Limit 10
Router1
Physical Config CLI Attributes
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Algorithm Settings
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RIP
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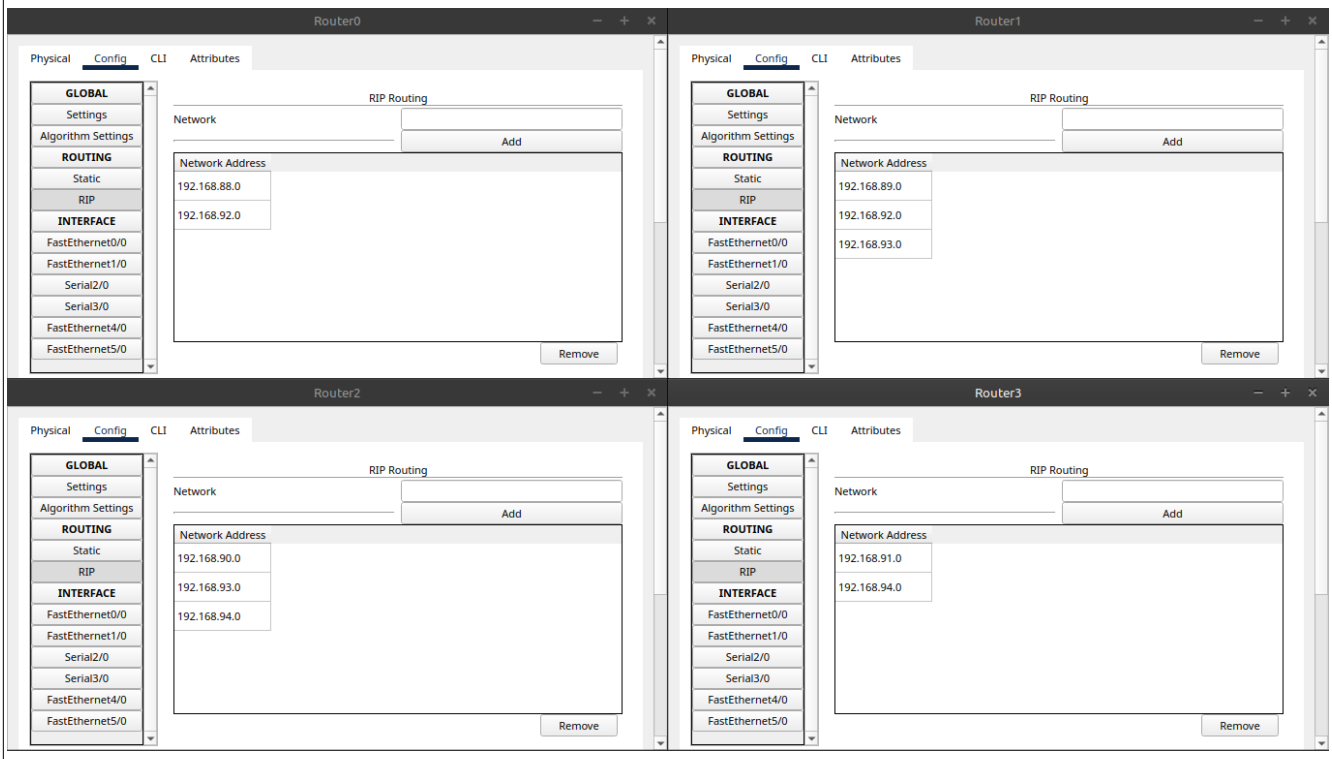
FastEthernet0/0
FastEthernet1/0
Serial2/0
Serial3/0
FastEthernet4/0
FastEthernet5/0
Serial3/0
Port Status On
Duplex Full Duplex
Clock Rate 2000000
IP Configuration
IPv4 Address 192.168.93.1
Subnet Mask 255.255.255.0
Tx Ring Limit 10
Router2
Physical Config CLI Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
INTERFACE
FastEthernet0/0
FastEthernet1/0
Serial2/0
Serial3/0
FastEthernet4/0
FastEthernet5/0
Serial3/0
Port Status On
Duplex Full Duplex
Clock Rate 2000000
IP Configuration
IPv4 Address 192.168.94.1
Subnet Mask 255.255.255.0
Tx Ring Limit 10
Router3
Physical Config CLI Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
INTERFACE
FastEthernet0/0
FastEthernet1/0
Serial2/0
Serial3/0
FastEthernet4/0
FastEthernet5/0
Serial3/0
Port Status On
Duplex Full Duplex
Clock Rate 2000000
IP Configuration
IPv4 Address
Subnet Mask
Tx Ring Limit 10

5. Dynamic Routing (RIP)

Routing Information Protocol (RIP) was configured on all routers by adding the respective network IDs. This enabled exchange of routing information dynamically among routers.



6. Testing & Verification

Ping tests were performed between all devices in the network. Successful transmission verified correct configuration and functioning of dynamic routing across the entire network.

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num
	Successful	Router0	Router1	ICMP		0.000	N	0
	Successful	Router0	Router2	ICMP		0.000	N	1
	Successful	Router0	Router3	ICMP		0.000	N	2
	Successful	PC1	Router0	ICMP		0.000	N	3

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num
	Successful	PC0	Router0	ICMP		0.000	N	4
	Successful	PC2	Router0	ICMP		0.000	N	5
	Successful	PC3	Router0	ICMP		0.000	N	6
	Successful	Router2	PC2	ICMP		0.000	N	7

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num
	Successful	Router2	PC3	ICMP		0.000	N	8
	Successful	Router3	Laptop0	ICMP		0.000	N	9
	Successful	Router3	PC1	ICMP		0.000	N	10
	Successful	Router1	Laptop0	ICMP		0.000	N	11

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num
	Successful	PC0	Laptop3	ICMP		0.000	N	12
	Successful	PC3	Laptop0	ICMP		0.000	N	13
	Successful	Laptop2	PC1	ICMP		0.000	N	14
	Successful	Laptop1	PC2	ICMP		0.000	N	15

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num
	Successful	Router3	PC1	ICMP		0.000	N	16
	Successful	Router3	Laptop2	ICMP		0.000	N	17
	Successful	Router3	Router1	ICMP		0.000	N	18
	Successful	Router1	Laptop3	ICMP		0.000	N	19

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num
	Successful	Router2	PC1	ICMP		0.000	N	20
	Successful	Router0	Laptop2	ICMP		0.000	N	21
	Successful	Router3	PC0	ICMP		0.000	N	22
	Successful	Router1	Router2	ICMP		0.000	N	23

7. Conclusion

The lab successfully demonstrated dynamic routing using RIP in a multi-router environment. Network communication worked effectively across all connected devices.